



Exam :

End Term Quiz

Subject :

Maths2

Total Marks :

50.00

QP :

2023 Sep03: IIT M DIPLOMA ET1 EXAM 2023 Sep03: IIT

Exam Mode

Learning Mode

View Question Paper Summary

QUESTION MENU

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CONTROLS

SUBMIT EXAM

Your Score

0.00 / 50.00

(0%)

Question 1 : 640653608650

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**FOUNDATION LEVEL : MATHEMATICS FOR DATA SCIENCE 2 (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET

TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ YES

☐ NO

Your score : 0

 Discussions (0)



Question 2 : 640653608671

 View Solutions (0)

Total Mark : 2.00 | Type : MCQ

A skier is on a mountain with equation $z = 20 - 0.4x^2 - 0.3y^2$ where z denotes the height. The skier is located at the point with xy -coordinates $(1, -1)$, and wants to ski downhill along the steepest possible path. In which direction should the skier begin skiing?



OPTIONS :

☐ (0.8, 0.6)

☐ (-0.8, 0.6)

☐ (-0.8, -0.6)

☐ (0.8, -0.6)

Your score : 0

 Discussions (0)



Question 3 : 640653608651

 View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Which of the following statement(s) is/are true for an $n \times n$ matrix A ?



OPTIONS :

☐ Let $\det(A) \neq 0$. Then determinant of A is unaltered by swapping any two rows of A .



☐ Determinant of A is unaltered when a multiple of a row is added to another row.

☐ For any real number t , $\det(tA) = t^n \det(A)$

☐ If $D = P^{-1}AP$, then $D = A$.

Your score : 0

Discussions (0)



Question 4 : 640653608652

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Which of the following statement(s) is/are true for a system of linear equations $Ax = b$?

OPTIONS :

☐ If x_1 and x_2 are solutions of the system, then for any $\alpha, \beta \in \mathbb{R}$, $\alpha x_1 + \beta x_2$ is also a solution if and only if $b = 0$.

☐ If x_1 and x_2 are solutions of the linear equation, then $\alpha x_1 + \beta x_2$ is also a solution for any b in the column space of A .

☐ The system $Ax = 0$ has a unique solution when the number of equations is less than the number of variables.

☐ If the system $Ax = 0$ has a unique solution, then the columns of A are linearly independent.

Your score : 0

Discussions (0)



Question 5 : 640653608653

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Let V be the vector space of all $n \times n$ matrices with usual addition and scalar multiplication. Which of the following set(s) form a subspace of V ?

OPTIONS :

☐ $W_1 = \{A \in M_n(\mathbb{R}) : A \text{ is invertible}\}$

☐ $W_2 = \{A \in M_n(\mathbb{R}) : \det(A) = 1\}$

☐ $W_3 = \{A \in M_n(\mathbb{R}) : A \text{ is upper triangular}\}$

☐ $W_4 = \{A \in M_n(\mathbb{R}) : A \text{ is symmetric i.e., } A^T = A\}$

Your score : 0

Discussions (0)



Question 6 : 640653608664

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Let $U = \{(x, y, z) \in \mathbb{R}^3 : x = y = z\}$ and $V = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. Let P_U and P_V be the projections on the spaces U and V respectively. Which of the following statement(s) is/are true?

OPTIONS :

☐ $\{(-1, 1, 0), (-1, -1, 2)\}$ is a basis for the range space of P_U .

☐ $\{(-1, 1, 0), (-1, -1, 2)\}$ is a basis for the null space of P_U .

☐ $\{(1, 1, 1)\}$ is a basis for the range space of P_V .

☐ $\{(1, 1, 1)\}$ is a basis for the null space of P_V .

Your score : 0

Discussions (0)



Question 7 : 640653608666

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Let $W = \{(x, y, z) \in \mathbb{R}^3 : 2x - y - z = 3\}$. Then which of the following affine subspace(s) of $\mathbb{R}^3$ is/are same as the subspace W ?

OPTIONS :

☐ $(6, 0, 0) + U$ where $U = \{(x, y, z) \in \mathbb{R}^3 : 2x = y + z\}$.

☐ $(0, -3, 0) + U$ where $U = \{(x, y, z) \in \mathbb{R}^3 : y = 2x - z\}$.

☐ $(0, 0, -3) + U$ where $U = \{(x, y, z) \in \mathbb{R}^3 : z = 2x - y\}$.

☐ $(0, -3, 0) + U$ where $U = \{(x, y, z) \in \mathbb{R}^3 : 2x = y - z\}$.

Your score : 0

Discussions (0)



Question 8 : 640653608673

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Let $f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{otherwise.} \end{cases}$. Then which of the following statement(s) is/are true?

OPTIONS :

- ☐ The partial derivatives for f exist everywhere but directional derivatives neednot exist at the origin.
- ☐ All directional derivatives at the origin exist for the function f .
- ☐ At $(0, 0)$, the directional derivative of f exists in the direction of x -axis.
- ☐ The partial derivatives for f do not exist at the origin.

Your score : 0

Discussions (0)



Question 9 : 640653608654

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $E = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ and $F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Then which of the following statement(s) is/are true?

OPTIONS :

- ☐ $\text{span}\{B, C, D\} = M_2(\mathbb{R})$.
- ☐ $\text{span}\{A, B, D\} = \text{span}\{A, F\}$
- ☐ $\text{Span}\{A, E, F\} = \{A \in M_2(\mathbb{R}) : A \text{ is symmetric i.e., } A^T = A\}$.
- ☐ The set of all 2×2 diagonal matrices are spanned by the matrices A and E .

Your score : 0

Discussions (0)



Question 10 : 640653608661

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Which of the following statement(s) about similar matrices is/are true?

OPTIONS :

- ☐ Let $T: V \rightarrow W$ be a linear transformation. If A and B are two different matrices corresponding to T , then A and B must be similar.
- ☐ If A and B are $n \times n$ matrices such that $\det(A) = \det(B) = 1$, then A and B are similar matrices.
- ☐ If A is similar to identity matrix, then A is a scalar matrix.
- ☐ If A is invertible, then AB and BA are similar.

Your score : 0

Discussions (0)



Question 11 : 640653608665

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Consider the following matrices.

$$A_1 = \begin{pmatrix} 2 & 2 \\ 0 & 2 \end{pmatrix}, A_2 = \begin{pmatrix} -1 & 2 \\ 3 & 1 \end{pmatrix}, A_3 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \text{ and } A_4 = \begin{pmatrix} -5 & -3 \\ 6 & 5 \end{pmatrix}$$

Choose the correct option(s) from the following:

OPTIONS :

- ☐ A_1 and A_3 are similar matrices
- ☐ A_1 and A_3 are equivalent but not similar matrices
- ☐ $A_4 = P^{-1}A_2P$ where $P = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$

☐ A_2 and A_4 are equivalent but not similar matrices 

Your score : 0

 Discussions (0)



Question 12 : 640653608663

 View Solutions (0)

Total Mark : 1.00 | Type : SA

Let A be an orthogonal matrix. Then the sum of squares of the elements of every row is :



Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

 Discussions (0)



Question 13 : 640653608662

 View Solutions (0)

Total Mark : 2.00 | Type : SA

Let u, v be vectors in $\mathbb{R}^3$ such that $u + v$ and $u - v$ are orthogonal. If $u = (1, -2, 2)$, then $\|v\| =$



Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

 Discussions (0)



Question 14 : 640653608672

 View Solutions (0)

Total Mark : 2.00 | Type : SA

Let $f(x, y, z) = x^2y^3 - 3xz$ and $u = (1, 2, 2)$. Find the directional derivative of f in the direction of the vector u at the point $(0, 1, -1)$.

Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

Discussions (0)



Question 15 : 640653608667

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let L_U denote an affine subspace with the associated subspace U of $\mathbb{R}^3$.

	Affine Subspace L_U		Associated subspace U		Dimension of L_U
(i)	$\{(x, y, z) \in \mathbb{R}^3 : x - y + z = -2, x + z = 1\}$	(1)	$\text{span}\{(1, 0, -1), (2, 1, 0)\}$	(a)	2
(ii)	$\{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 3\}$	(2)	$\text{span}\{(2, -1, 3)\}$	(b)	1
(iii)	$\{(x - 1, y + 1, z) \in \mathbb{R}^3 : x + 2y = 0, 3y + z = 0\}$	(3)	Nullspace of $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$	(c)	1

Table: M2ES2

Your score : 0



Question 16 : 640653608668

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Choose the correct option to match the affine subspace of Row 1 with the associated subspaces and dimension.

OPTIONS :

☐ (i) $\rightarrow$ (3) $\rightarrow$ (b)

- ☐ $(i) \rightarrow (2) \rightarrow (a)$ 
- ☐ $(i) \rightarrow (2) \rightarrow (c)$ 
- ☐ $(i) \rightarrow (1) \rightarrow (a)$ 

Your score : 0

 Discussions (0)



Question 17 :

640653608669



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Choose the correct option to match the affine subspace of Row 2 with the associated subspaces and dimension.

OPTIONS :

- ☐ $(ii) \rightarrow (3) \rightarrow (a)$ 
- ☐ $(ii) \rightarrow (1) \rightarrow (a)$ 
- ☐ $(ii) \rightarrow (2) \rightarrow (c)$ 
- ☐ $(ii) \rightarrow (1) \rightarrow (b)$ 

Your score : 0

 Discussions (0)



Question 18 :

640653608670



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Choose the correct option to match the affine subspace of Row 3 with the associated subspaces and dimension.

OPTIONS :

☐ $(iii) \rightarrow (3) \rightarrow (a)$

☐ $(iii) \rightarrow (2) \rightarrow (c)$

☐ $(iii) \rightarrow (2) \rightarrow (a)$

☐ $(iii) \rightarrow (3) \rightarrow (b)$

Your score : 0

Discussions (0)



Question 19 : 640653608674

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $f(x, y) = \sin x \cos y$.

Your score : 0



Question 20 : 640653608675

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Choose the correct option for the parametric equations of the line tangent to f at the point $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ in the direction of x axis:

☐ $x(t) = \frac{\pi}{2}, y(t) = \frac{\pi}{2} - t, z(t) = 0$

☐ $x(t) = \frac{\pi}{2} + t, y(t) = \frac{\pi}{2}, z(t) = 0$

☐ $x(t) = \frac{\pi}{2} + t, y(t) = \frac{\pi}{2}, z(t) = t$ 

Your score : 0

 Discussions (0)



Question 21 :
640653608676

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Choose the correct option for the parametric equations of the line tangent to f at the point $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ in the direction of y axis: 

OPTIONS :

☐ $x(t) = \frac{\pi}{2}, y(t) = \frac{\pi}{2} + t, z(t) = -t$ 

☐ $x(t) = \frac{\pi}{2} + t, y(t) = \frac{\pi}{2}, z(t) = t$ 

☐ $x(t) = \frac{\pi}{2}, y(t) = \frac{\pi}{2} - t, z(t) = 0$ 

Your score : 0

 Discussions (0)



Question 22 :
640653608677

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Choose the correct option for the parametric equations of the line tangent to f at the point $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ in the direction of $(-1, 1)$: 

OPTIONS :

- ☐ $x(t) = \frac{\pi}{2} + \frac{t}{\sqrt{2}}, y(t) = \frac{\pi}{2} - \frac{t}{\sqrt{2}}, z(t) = -\frac{1}{\sqrt{2}}$ 
- ☐ $x(t) = \frac{\pi}{2} - \frac{t}{\sqrt{2}}, y(t) = \frac{\pi}{2} + \frac{t}{\sqrt{2}}, z(t) = \frac{t}{\sqrt{2}}$ 
- ☐ $x(t) = \frac{\pi}{2} - \frac{t}{\sqrt{2}}, y(t) = \frac{\pi}{2} + \frac{t}{\sqrt{2}}, z(t) = -\frac{t}{\sqrt{2}}$ 

Your score : 0

 Discussions (0)



Question 23 : 640653608655

Total Mark : 0.00 | Type : COMPREHENSION

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and $A = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ be the matrix corresponding 
to the linear transformation T with respect to the bases
 $\beta = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and $\gamma = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$.
Using the above information answer the given subquestions.

Your score : 0



Question 24 :

640653608656

 View Parent QN

 View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Choose the appropriate linear transformation T for the matrix A .

OPTIONS :

- ☐ $T(x, y, z) = (x + y, y + z, z - x)$ 
- ☐ $T(x, y, z) = (x, y - z, z + x)$ 

☐ $T(x, y, z) = (x + y, y - z, z + x)$

☐ $T(x, y, z) = (x + y, -z, z + x)$

Your score : 0

Discussions (0)



Question 25 :

640653608657

View Parent QN

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let B be the matrix corresponding to the linear transformation T with respect to the basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ for both domain and codomain. Which of the following statement(s) is/are true?

OPTIONS :

☐ The column space of B is spanned by $\{(1, 0, 1), (1, 1, 0)\}$

☐ The nullspace of B is spanned by $\{(-1, 0, 0), (0, 1, 1)\}$

☐ $\text{rank}(B) = 2$ and $\text{nullity}(B) = 1$

☐ $\text{rank}(B) = 1$ and $\text{nullity}(B) = 2$

Your score : 0

Discussions (0)



Question 26 :

640653608658

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : SA

Let B be the matrix corresponding to the linear transformation T with respect to the basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ for both domain and codomain. The trace of B is

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

 Discussions (0)

Question 27 :

640653608659



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Let B be the matrix corresponding to the linear transformation T with respect to the basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ for both domain and codomain. The determinant of B is

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

 Discussions (0)

Question 28 :

640653608660



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Which of the following statement(s) about T is/are true?

OPTIONS :

☐ T is one-one but not onto ☐ T is neither one-one nor onto. ☐ Matrices corresponding to T with respect to any basis has determinant zero. 

- ☐ There exists a basis β of $\mathbb{R}^3$ such that the matrix corresponding to T with respect to β for both domain and codomain is invertible.

Your score : 0

 Discussions (0)



 SUBMIT EXAM